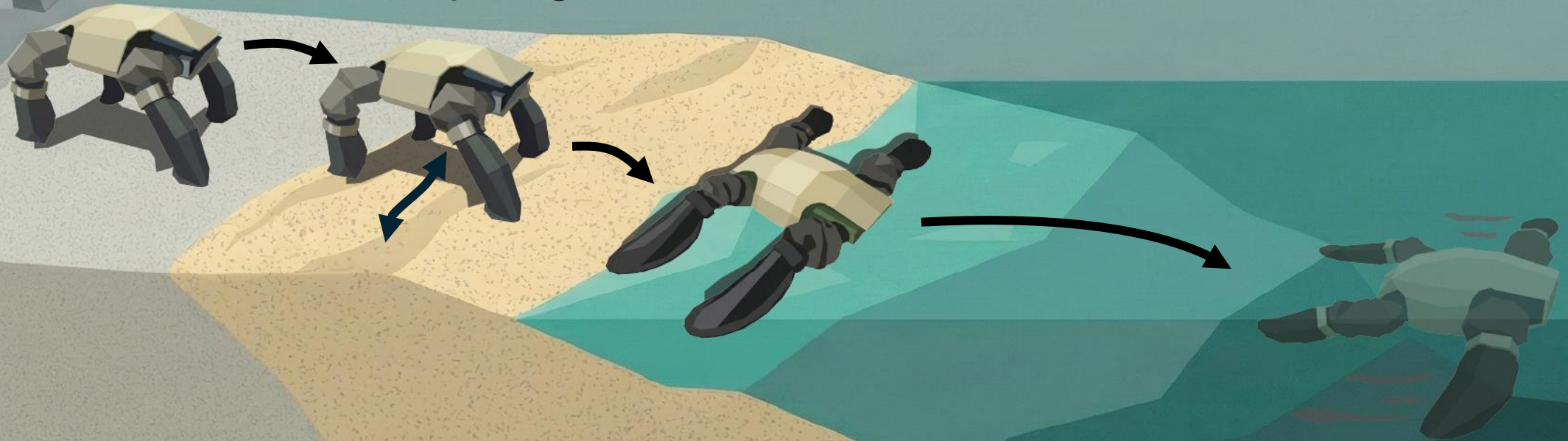


Full-Envelope Amphibious Locomotion

with a Morphing Robotic Turtle



A morphing amphibious robot

Learning Untethered Amphibious Robotic Turtle (LUART)



Leg mode

inflated, stiff limbs



Flipper mode

deflated, flat limbs

Active morphing

4x



01 On land

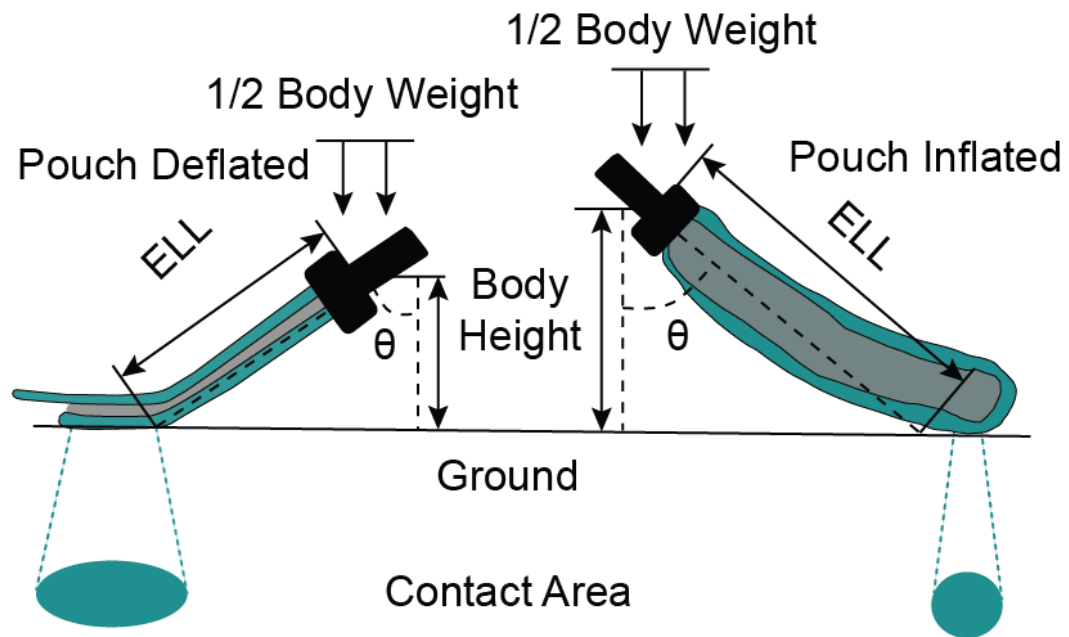
02 In water

03 At the transition

Softness surrogates in rigid simulation

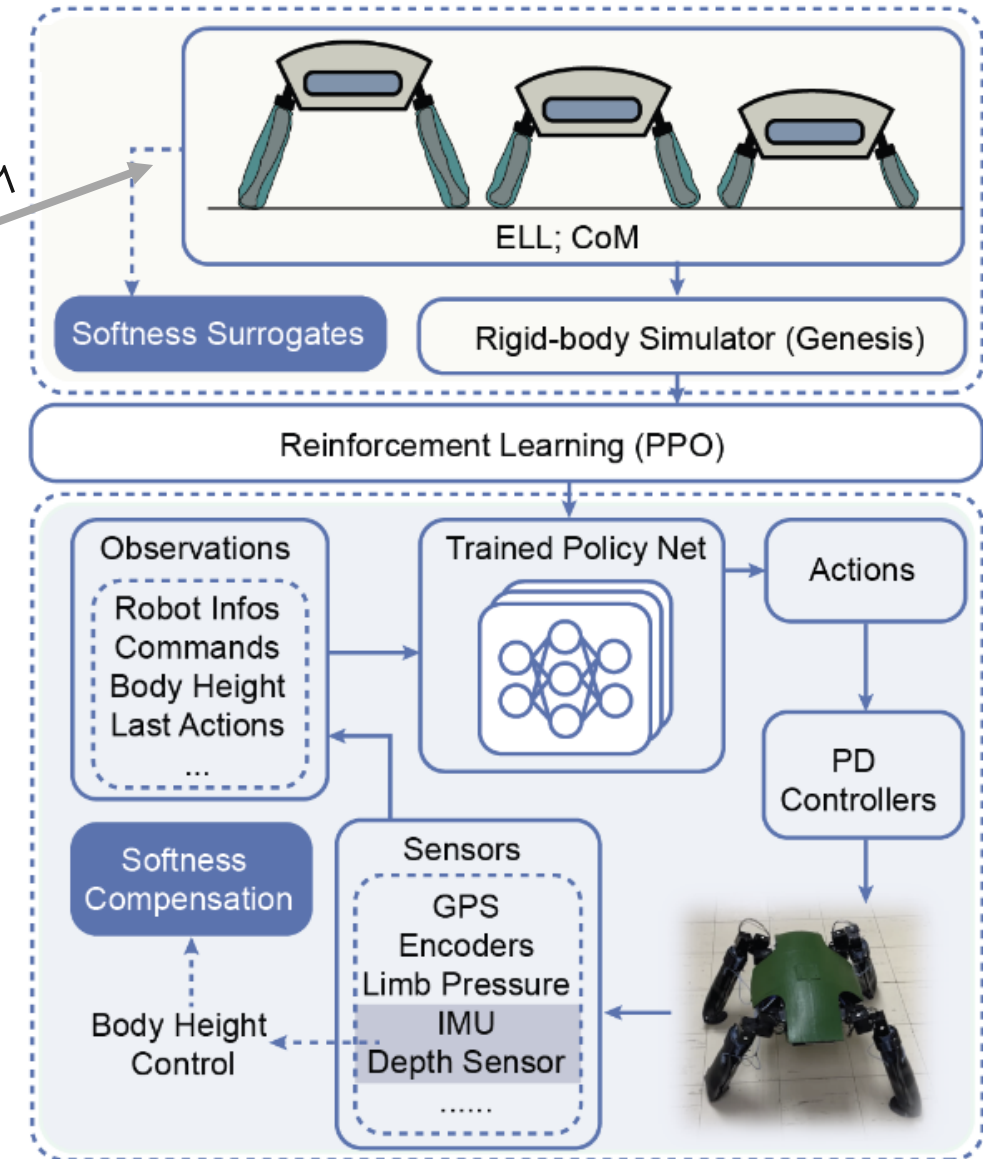
01 ON LAND

Encode soft limbs in a rigid simulator



0.5–1.1

Softness Surrogate: Effective leg length, friction
Softness Compensation: Body height control



Agile and adaptive locomotion

01 ON LAND

Agility

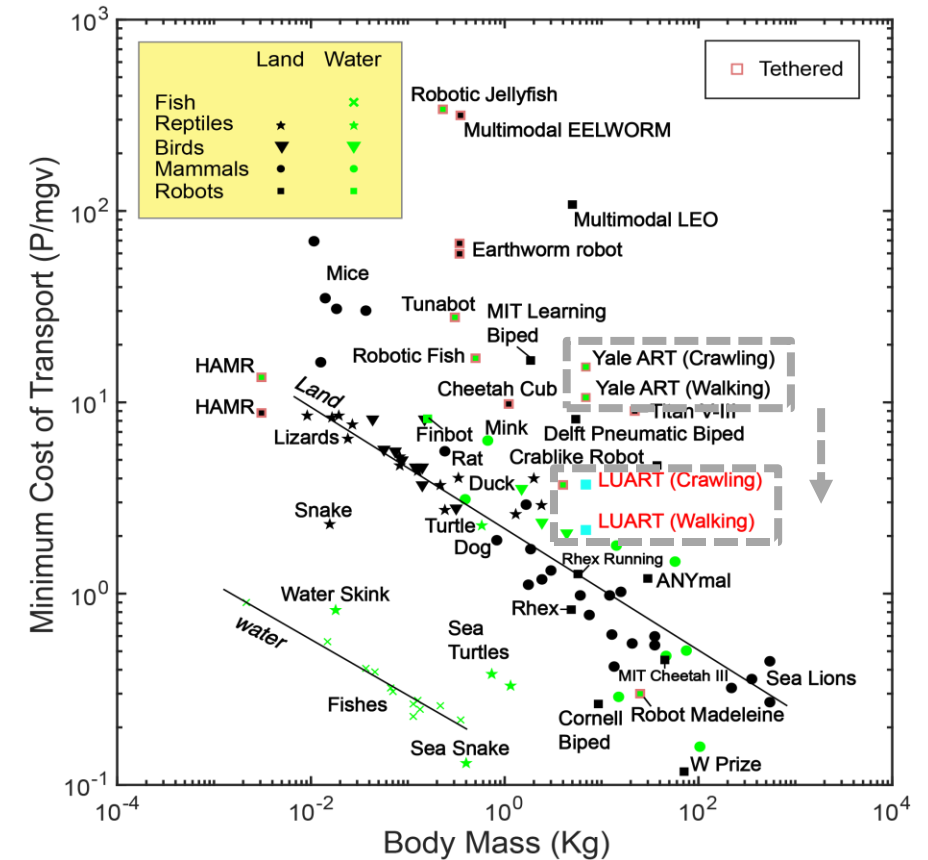
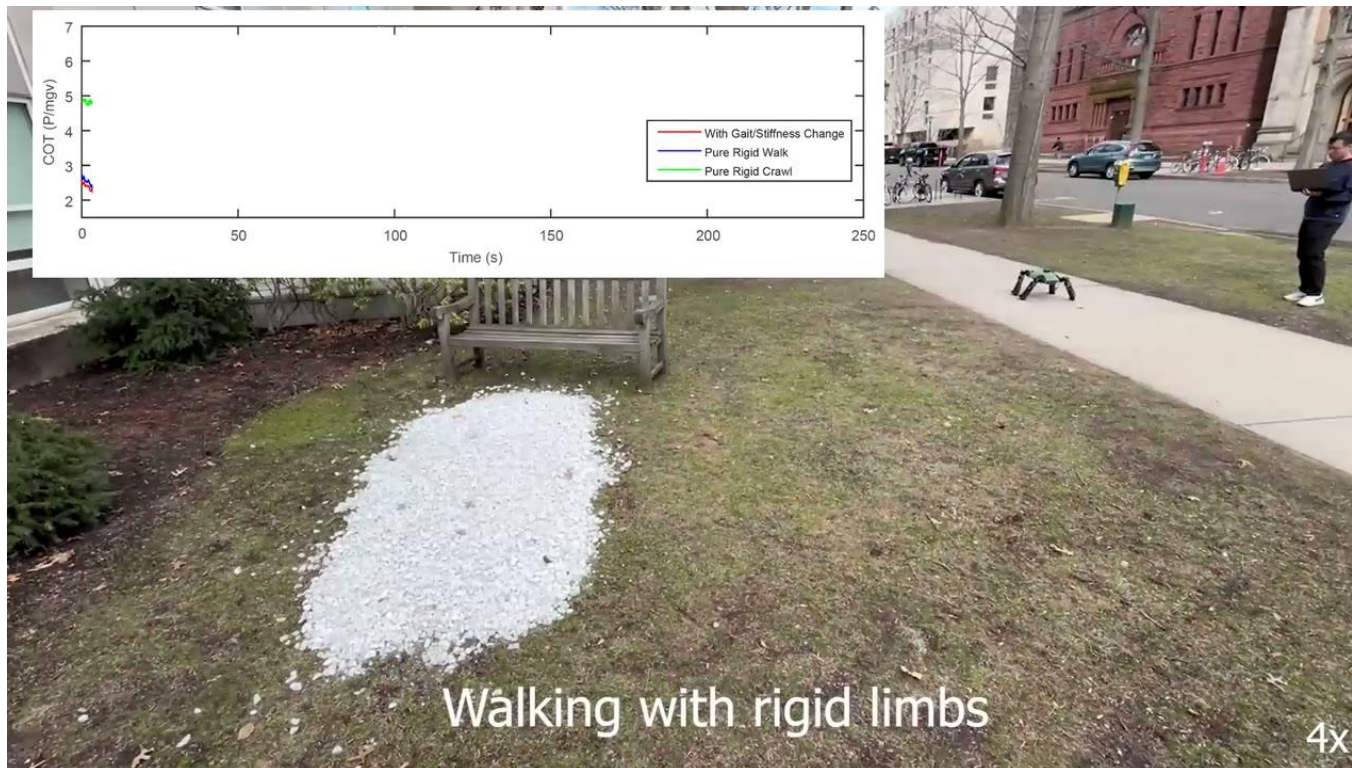


Adaptivity



Energy efficiency with real-time COT

01 ON LAND

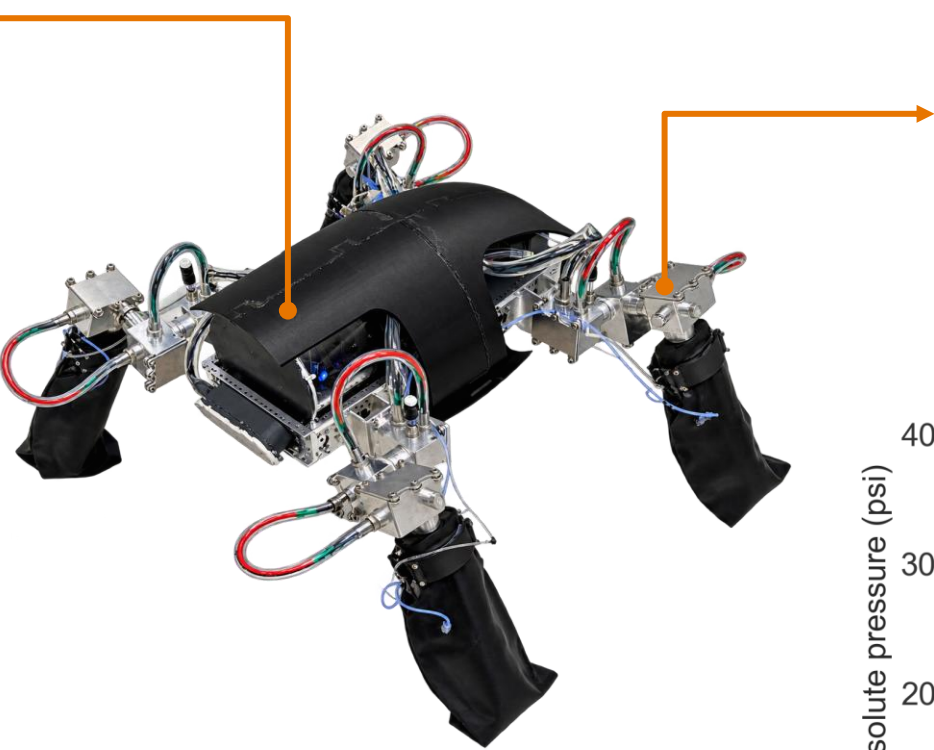
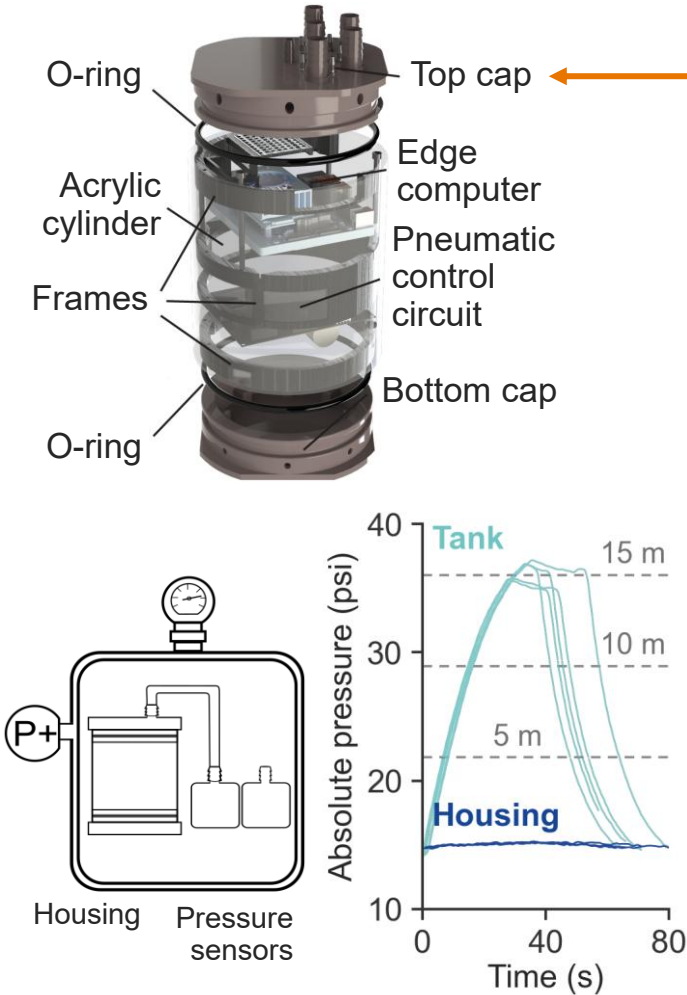


Waterproof hardware for 10 m depth

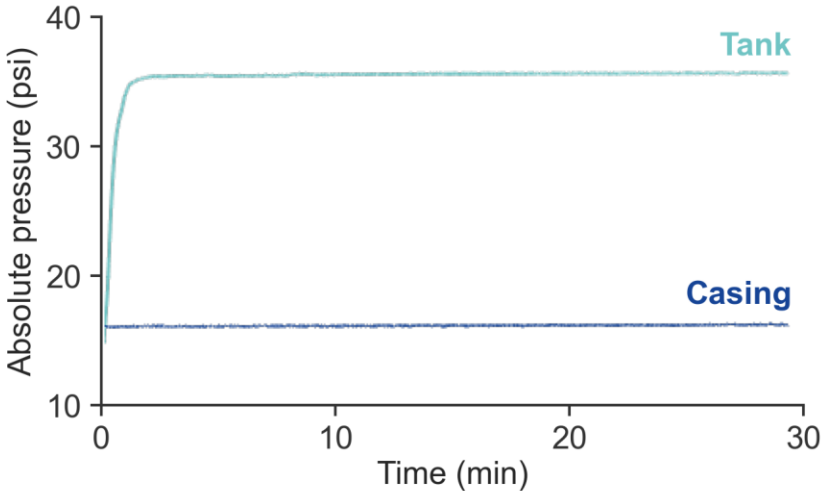
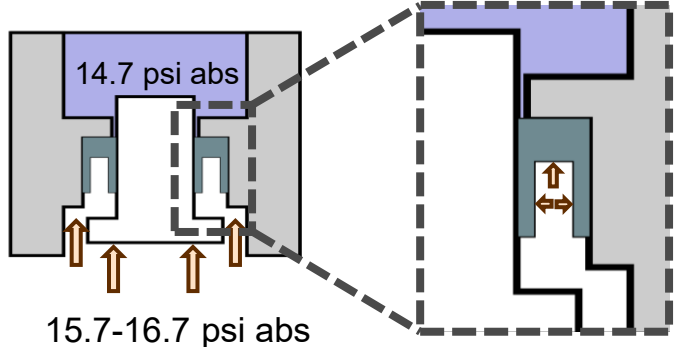
Controller housing

Pressure Regulation

Motor casing



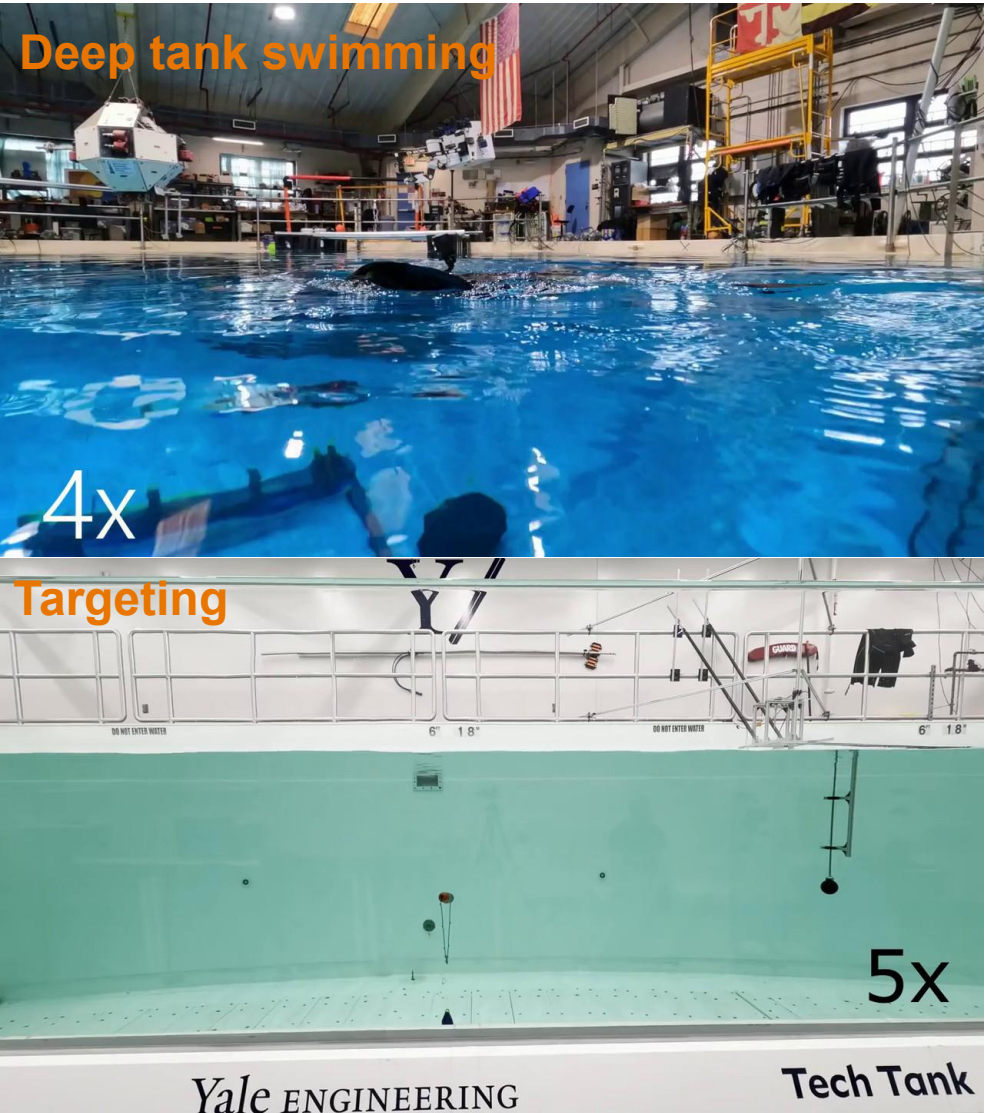
Inside kept 1–2 psi above outside
Positive pressure keeps water out, from the surface to 10 m



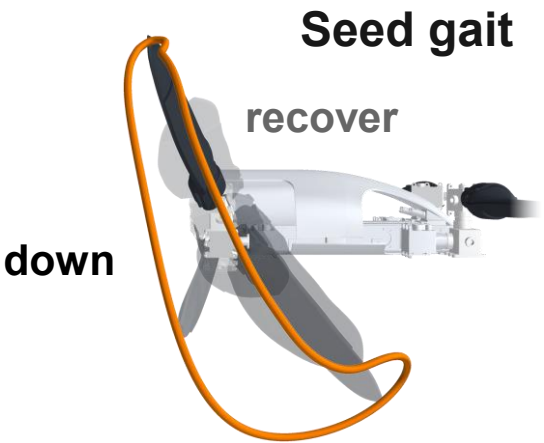
Swimming deep and agile / Learning gaits

02 IN WATER

Hand-designed gaits



Learned gaits



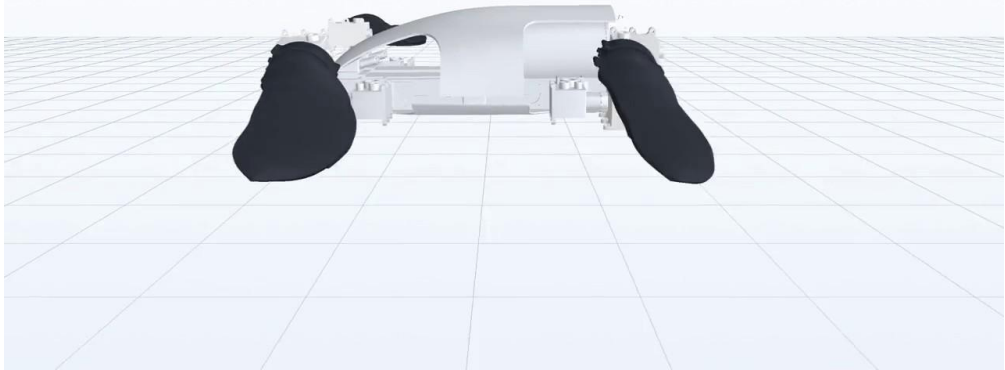
Work in progress

- 1 Run on the real robot
- 2 Calibrate ellipsoid model to robot data
- 3 RL training from the seed gait
- 4 3D attitude and speed control

Swim faster

speed command 0.20 m/s robot 0.13 m/s

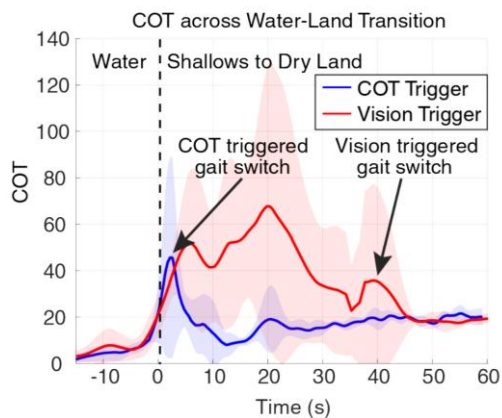
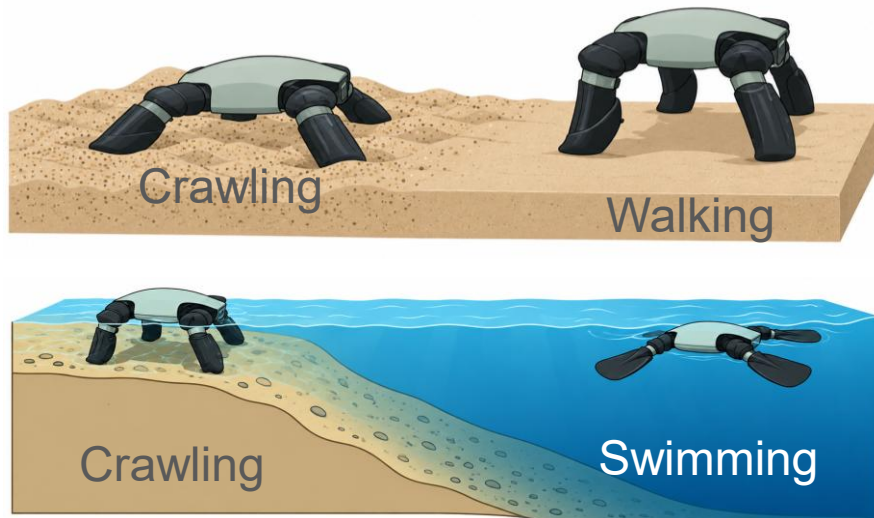
Ellipsoid model 4x



Energy-aware transitions: MEPG planning

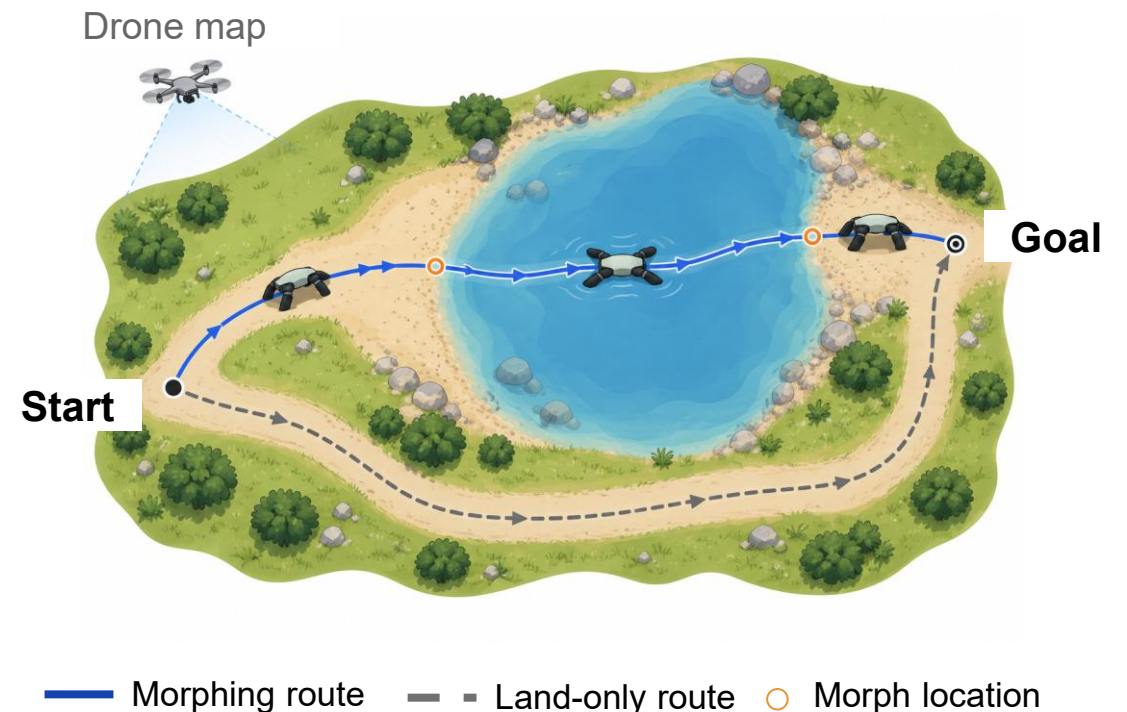
03 AT THE TRANSITION

Real-time COT trigger



Morph–Energy–Path–Gait planning

Work in progress



Minimum total energy, including transitions

Thank you

Jue Wang · Princeton University · wang5056.github.io